

MICROBIOLOGY

CHAPTER 21 : GRAM POSITIVE



**Done by:
Sundos ali**

**Reference:
Slide and
Dr sereen notes**

Chapter 21: Gram positive bacteria

→ After gram staining and finding gram +ve bacteria we will only find two shapes:

Cocci & Bacilli

*Spiral shaped bacteria
is only gram -ve*

• Peptidoglycan structure of gram positive bacteria

- Gram +ve cell wall is a thick layer of peptidoglycan.
- Peptidoglycan chains are linked by a peptide interbridge which distinguishes it from gram -ve (the photo on the right)
- The reason why the structure of peptidoglycan is important in classification is that a small change in the linkage gives a whole different type of bacteria $\Sigma(O_2)$

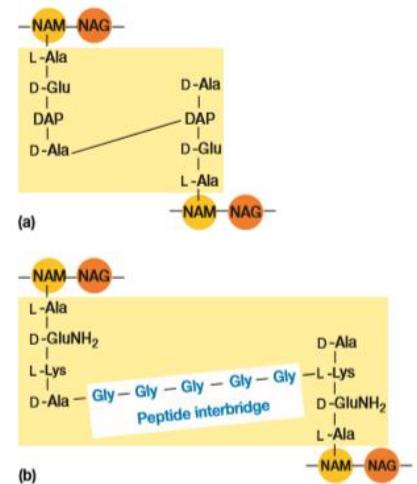
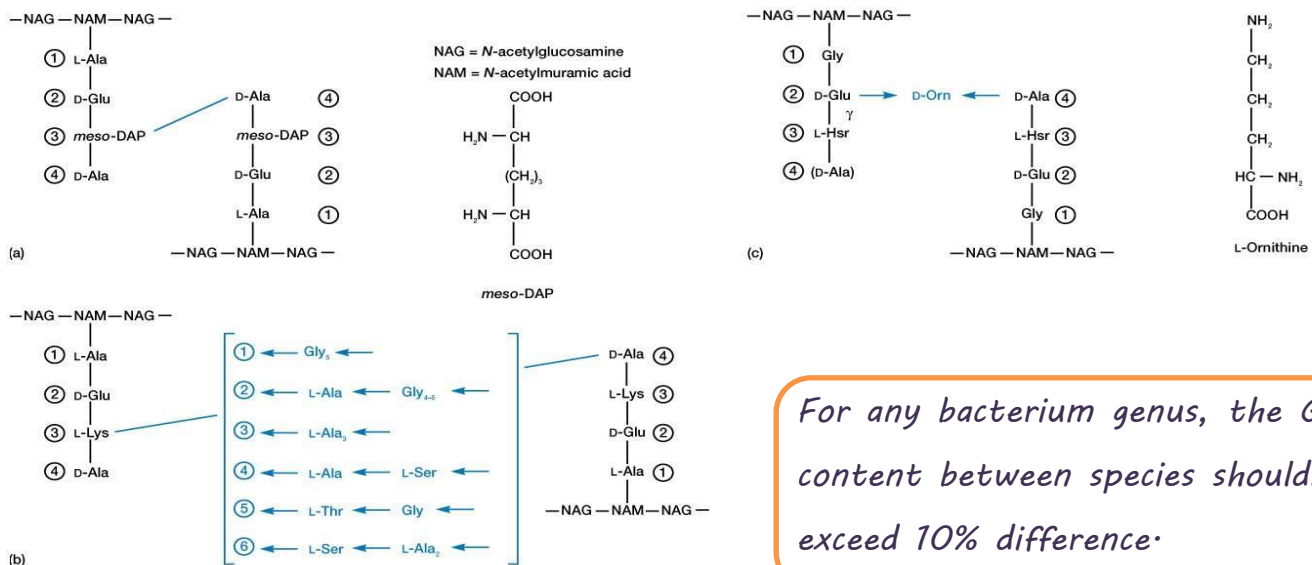


Figure 3.19 Peptidoglycan Cross-Links Can Be Direct or Indirect via a Peptide Interbridge. (a) *E. coli* peptidoglycan with direct cross-linking, typical of many Gram-negative bacteria. (b) *Staphylococcus aureus* peptidoglycan with an interbridge. *S. aureus* stains Gram positive. NAM is N-acetylmuramic acid. NAG is N-acetylglucosamine. Gly is glycine. D-GluNH₂ is D-glutamic acid with an NH₂ group attached to the α carbon (the carbon next to the carboxyl group).

كمثال، خلينا نقول انو الرابطة (الجسر) الي بين ال meso-DAP وال D-Ala (الحمض الاميني الثالث من السلسلة الاولى مع الحمض الاميني الرابع من السلسلة الثانية) بالفرع (a) تغيرت وبدل ما تكون بينهم صارت بين ال meso-DAP وال L-Ala (الحمض الاميني الثالث من السلسلة الاولى مع الحمض الاميني الاول من السلسلة الثانية) فهاد التغير رح يعطيني نوع جديد ومختلف تماما.



For any bacterium genus, the G-C content between species shouldn't exceed 10% difference.

I. Cocci shaped bacteria (NONmotile)

a) Aerobic:

1. Micrococcus
2. Deinococcus

→ Catalase Positive

b) Facultative Anaerobic:

1. Staphylococcus
2. Streptococcus
3. Enterococcus
4. Lactococcus
5. Leuconostoc

c) Anaerobic:

Sarcina

→ Catalase Negative

Remember: Only Aerobic bacteria have the catalase enzyme since they produce oxygen singlates and they need the catalase enzyme in order to break down H_2O_2 into O_2 and H_2O

a) 1. Micrococcus:

- Aerobic, Catalase-positive rods that occur in pairs, tetrads or irregular clusters
- usually nonmotile
- often pigmented yellow, orange or red to protect the bacteria from photo lethal oxidation caused by visible light.

الاطوال الموجية المختلفة للضوء بتأثر بشكل سلبي على البكتيريا بشكل عام، حتى الضوء المرئي كثير منو بأثر عليها ويعملها أكسدة ضوئية مميتة. الصبغات الي بتكونها البكتيريا بتحميها من هاي الاكسدة.

- widespread in soil, water, and on human skin
- does not undergo morphological differentiation
- Micrococcus (oxidase +) and staphylococcus (oxidase -) have a lot in common (like colony morphology), the major difference between the two of them is their oxygen requirement.

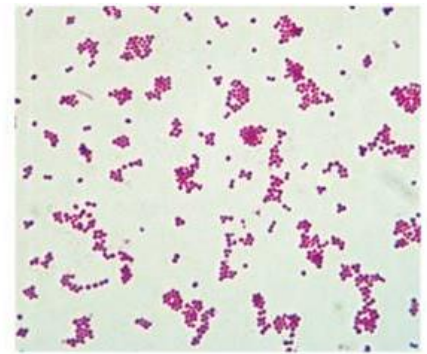
- التشابه بين ال Micrococcus وال Staphylococcus كثير كبير ويمكن يوصل ل 90% . الفرق الاساسي بينهم هو حاجتهم للاكسجين، ال Micrococcus هوائية وال Staphylococcus لا هوائية وعشان هيك لما نعملهم oxidase test ال Micrococcus بتطلع +ve وال Staphylococcus بتطلع -ve .

2. *Deinococcus*

- *Deinococcus radiodurans*: have S layer ,highly resistant to radiation

b) 1. Family *Staphylococcaceae*

- contains 4 genera
- facultatively anaerobic, nonmotile, gram-positive cocci
- usually form irregular clusters (See the figure)
- normally associated with warm blooded animals in skin, skin glands and mucous membranes
- *Staphylococcus* produces golden colonies when grown on petri dishes



(a)



(b)

© Bruce Iverson
© Photo Researchers, Inc.

Pathogenic *Staphylococcus*

Staphylococcus
epidermidis

Staphylococcus
aureus

- *Staphylococcus epidermidis*

- common skin resident
- sometimes responsible for endocarditis (التهاب داخلي بالقلب) and for infections of patients with lowered resistance
e.g., wound infections, surgical infections, and urinary tract infection

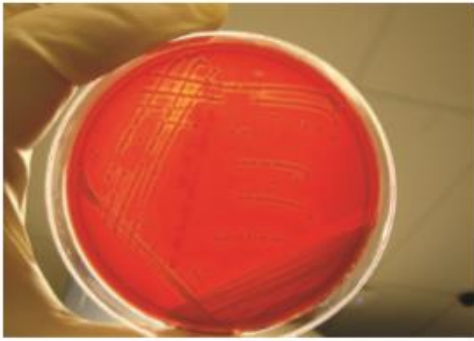
➤ Blood Hemolysis : Two types

α -hemolysis

- incomplete lysis of red blood cells
- seen as greenish zone around colony on blood agar

β -hemolysis

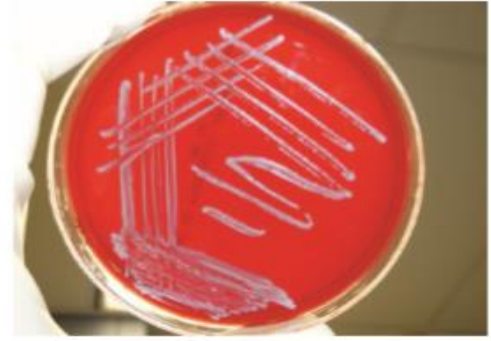
- complete lysis of red blood cells
- seen as clear zone around colony on blood agar



(a) Alpha-hemolysis



(b) Beta-hemolysis



(c) No hemolysis

Figure 23.15 Staphylococcal and Streptococcal Hemolytic Patterns. (a) *Staphylococcus aureus* on blood agar, illustrating α -hemolysis. (b) *Streptococcus pyogenes* on blood agar, illustrating β -hemolysis. (c) *Staphylococcus epidermidis* on blood agar with no hemolysis, also referred to as γ -hemolysis.

- *Staphylococcus aureus*

- found on nasal membranes and skin, and in gastrointestinal and urinary tracts
- produces α -hemolysin :toxin which lyses cells

بما انها بتفرز α -hemolysin فهي بتعمل α -hemolysis يعني هاد التوكسين هو اللي بعمل تحلل جزئي لخلايا الدم

- produces the virulence factor **Coagulase** enzyme

→ produces a clot of blood plasma to hide in it since its "normal" for the immune system to have them, and when It's safe it goes out and causes infection

من الطبيعي للجسم انو يكون فيه تكتلات لبلازما الدم يعني انها لا تُهاجم من قبل الجهاز المناعي، ف هاي البكتيريا بتفرز **Coagulase** اللي بعمل هاي التكتلات لتتخبي البكتيريا جواها وبتنتقل بالدم لأماكن تانية حتى يصير الوسط مناسب وآمن لها لتعيش فيه، بتطلع من هاد التكتل وبتسبب infection (٧٧٧)٦

- causes acne (pimples), wound infection and it's a major cause of food poisoning
 - recently >1,000 school children in Texas had staphylococcal food poisoning caused by eating improperly handled chicken
- A very important thing about *Staphylococcus aureus* is that it develops resistant → (Methicillin-resistant *Staphylococcus aureus* (MRSA)) → obtained from genetic elements received from other organisms
- Its even resistant to **Vancomycin**, the "drug of last resort"

لما ولا مضاد حيوي يزيط بلجأو لل **Vancomycin** على انو اقوى مضاد حيوي ، ولكن حتى هاد ال *Staphylococcus* طورت مقاومة ضده (° ° °)

- If we are preparing a selective media special for staphylococcus , it has to have a 7.5% salt concentration

→ *Staphylococcus aureus* and *staphylococcus epidermidis* are common residents on the skin

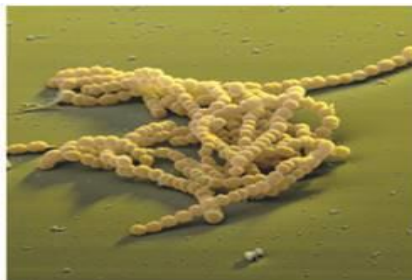
2. *Streptococcus*

- facultative anaerobic gram + cocci and catalase -ve
- Their major habitat is the mouth and the respiratory tract so based on that they are divided into two groups:

Group	Blood hemolysis	Species	infection
Oral	α - hemolysis	<i>Streptococcus orialis</i>	They cause Gingivitis (التهاب اللثة), tooth decay (تسوس الاسنان) and oral infections
		<i>Streptococcus mutans</i>	
		<i>Streptococcus pneumoniae</i>	lobar pneumonia and otitis media (this one does not have the bluish color found in pseudomonas)
Pyogenic	β - hemolysis	<i>Streptococcus pyogenes</i>	-streptococcal sore throat (التهاب الحلق) -Acute glomerulonephritis (التهاب كبيبات الكلى) -Rheumatic fever (حمى الروماتيزم) -tonsillitis (التهاب اللوز)



(a) *Streptococcus pyogenes*



(b) *S. agalactiae*
© Thomas Tottleben/Tottleben Scientific Company
© Photo Researchers, Inc.
© M. Abbey/Visuals Unlimited



(c) *S. pneumoniae*

- We do a catalase test to differentiate between *Staphylococcus* (catalase +ve) and *Streptococcus* (catalase -ve), because they are almost similar in shape when seen under microscope.

3. *Enterococcus*

- Found in the gastrointestinal tract
- They do α -hemolysis but then change to β -hemolysis
بمراحل حياتهم المختلفة يختلف نوع blood hemolysis يعني ممكن يكونوا بالاول الفا بعدين يتغيروا لبيتا
- Pollution indicator
- *Enterococcus faecalis* :
 - They are opportunistic meaning that if they had the chance to move to other tissues, they would and when they move to the urinary tract they cause urinary infection
 - They cause endocarditis

4. *Lactococcus*

- They are found in Dairy products
- **Homolactic** → they only one type of acid
→ lactic acid
- *Lactococcus lactis* - production of buttermilk and cheese by fermenting lactose
- has a special cell wall that activates lymphocytes thus protecting from colon cancer

Remember that any bacterial name that have "lacto", it means that this bacteria ferments milk into cheese and yogurt.

ولكن هاد لا يعني انو كل بكتيريا بتعمل تخمر للحليب
يعني لازم يكون باسمها *lacto* ، في منهم بعملوا تخمر
وما في باسمهم هاد المقطع

5. *Leuconostoc*

- facultative, gram-positive cocci
- isolated from plants, silage and milk
- play a role in food spoilage
 - tolerate high sugar concentrations
 - grow in heavy syrup
- carry out **Heterolactic** fermentation via phosphoketolase pathway
→ **Heterolactic fermenters** → more than one product → different types of acids



From M.P. Starr, et al. (Eds.), The Prokaryotes, Springer-Verlag

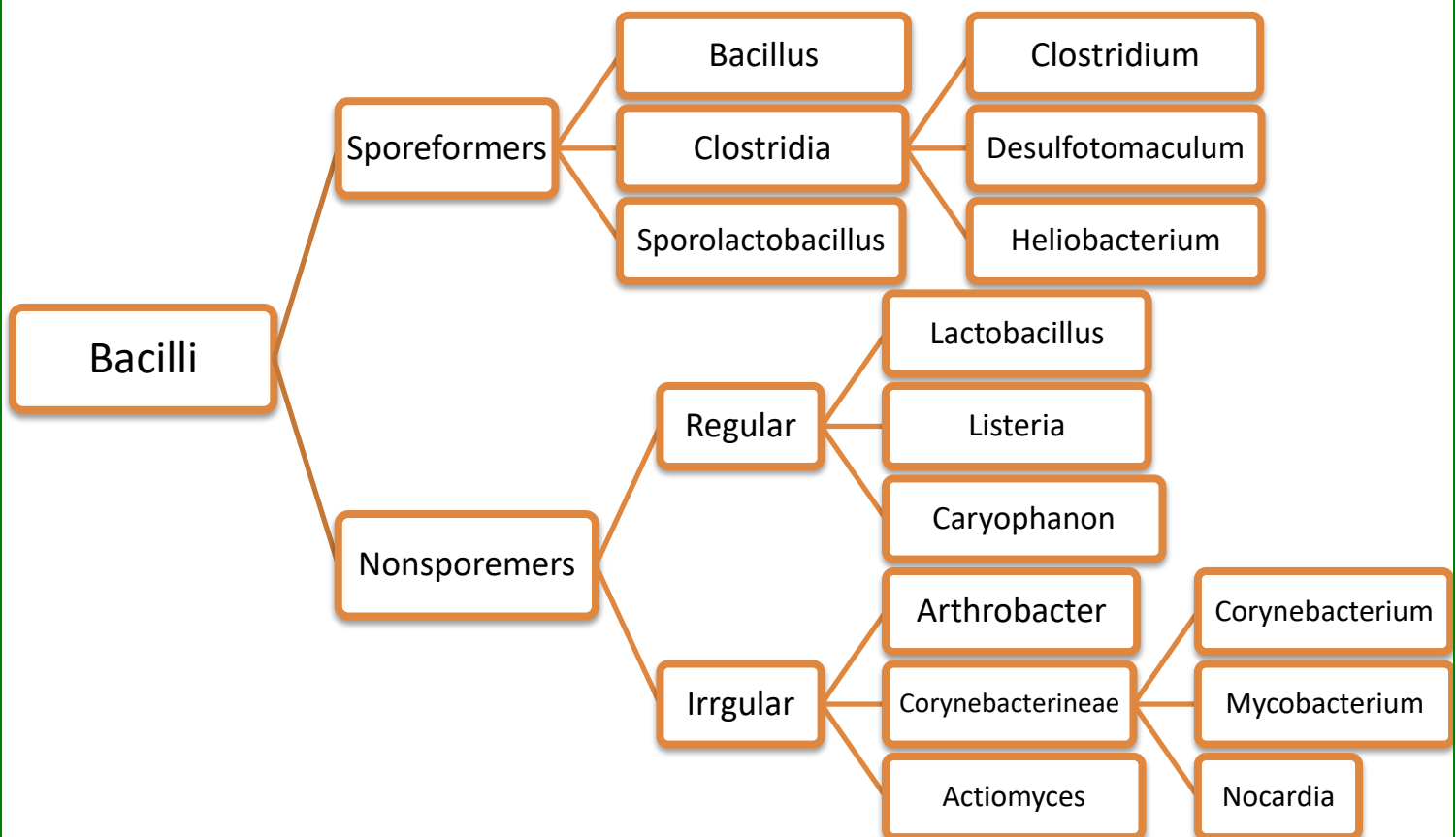
- Importance of Leuconostoc:
 - wine production
 - production of sauerkraut and pickles
 - production of buttermilk, butter, and cheese
 - (L. mesenteroides) :synthesis of **Dextrans**:
used medically to reduce blood viscosity.

** (any bacteria that do fermentation usually produce dairy products and beverages)*

c) Sarcina

- anaerobic and catalase -ve since it does not make oxygen singlets.

II. Bacilli shaped bacteria

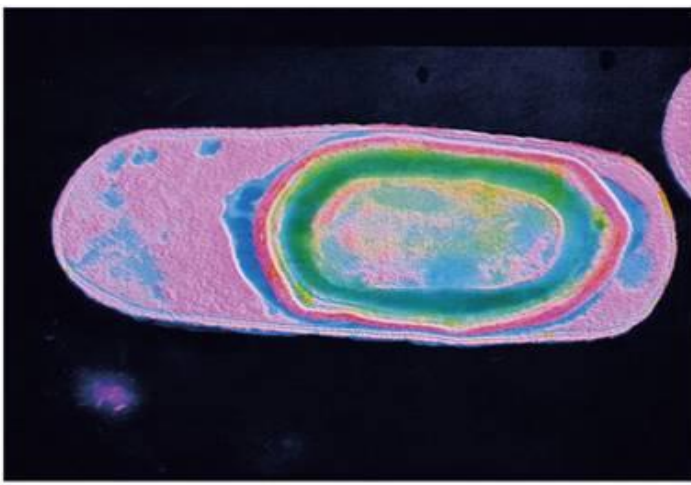


Endospores

- have a complex structure containing a coat, cortex, and inner spore membrane surrounding the protoplast
- heat resistant
- **Dipicolinic acid** is present and mainly responsible for heat resistancy of endospores.
- dormant and viable for long periods of time
- Spores need activation for turning into a vegetative cell, heat is the most important step in activation (providing nutrients without heat isn't enough for activation).

Remember : only gram positive bacteria produces endospores

تنشيط الـ endospores وإرجاعها لوضعها الطبيعي بصير بالحرارة.. بدون حرارة البكتيريا لا يمكن ترجع من endospore لـ vegetative cell حتى لو تم توفير الغذاء لـها.



(a)



(b)

CNRI/Photo Researchers, Inc.
© Dr. Tony Brain/Photo Researchers, Inc.

→ Based on spore forming, gram positive bacilli are divided into two groups :

- A) Sporeformers
- B) Nonsporeformers

The only gram positive genus that have **cocci** shaped bacteria that can form spores is *Sporosarcina* (aerobic and catalase +).

A. Sporeformers

1. Bacillus

- Class Bacilli

- large variety of gram-positive organisms
- contains two orders, Bacillales and Lactobacillales , 17 families and over 70 genera

The ONLY genus that is bacilli shaped, Aerobic and sporeformer is genus *Bacillus*

- *Bacillus subtilis*

- type species
- usually found in soil
- its genome was one of first to be sequenced
 - has families of genes expanded by gene duplication
 - ≥ 10 integrated prophages or remnants of prophages

لما عملوا sequencing للجينوم تبع ال *Bacillus subtilis* وجدوا عائلات من الجينات توسعت بفعل ال *gene duplication* بالإضافة الى انهم لقوا اكثر من 10 prophages في الجينوم تبعها.. ال prophage هو phage حقن المادة الوراثية تبعته ودمجها بالمادة الوراثية تبعت البكتيريا بس ضل بالخلية بدون ما يتكاثر.

- various species produce antibiotic

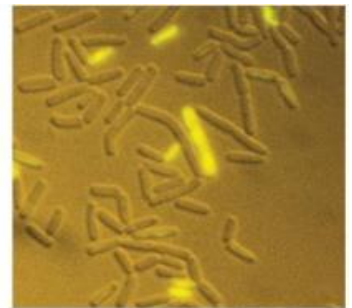
وهاي نقطة كثير مهمة. احنا منعرف انو لما اكتشو المضادات الحيوية اكتشفوها بالفطريات (*fungi*) ولكن هلا معظم المضادات الحيوية مأخوذة من بكتيريا ووحدة من هذول هي ال *Bacillus subtilis*

- Other important species of *Bacillus*

- *B. cereus* - food poisoning
- *B. anthracis* :
 - has a potential to be used as a bioweapon (letter attack 2001).
 - has a capsule that protects it from the immune system
 - it's a spore former (produces endospores)
 - causes anthrax (الجمرة الخبيثة).



(a) *Bacillus anthracis*



(b) *B. cereus*
© Arthur M. Siegelman/Visuals Unlimited
Courtesy of Molecular Probes, Inc.

- *B. thuringiensis* and *B. sphaericus* - used as insecticide
 - **Parasporal body** - solid protein crystal that contains toxin

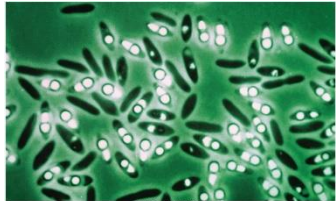
الـ parasporal body من *B. thuringiensis* و *B. sphaericus* يحتوي على endotoxin يمكن استخدامه كمبيد حشري بيولوجي أفضل بكثير من المواد الكيميائية (الـ toxin هو البكتيريا التي يتم استخدامه).

2. Class Clostridia

- contains three orders and 11 families
- Anaerobic and Catalase negative

Genera	Motility	Pathogenicity\Notes
Clostridium	Nonmotile or peritrichous	• over 100 species in distinct phylogenetic clusters ○ in future will probably be divided into several genera • fermentative metabolism ➤ ferment amino acids using Stickland reaction → oxidation of one amino acid using another as electron acceptor • fermentation products responsible for unpleasant odors associated with putrefaction (تعفن بالانسجة) <i>C. botulinum</i>: - food spoilage (especially canned foods) - a potential to be used as a bioweapon → causes botulism which is a fatal illness caused by a toxin produced by this bacteria. (This bacteria is so dangerous, it makes a highly powerful toxin (1 gram of it can kill up to 1 million people since its lethal dose is in nanograms))



		<i>C. tetani</i>: - causes tetanus (الخنق) (locked-jaw paralysis)  C. tetani © Arthur W. Englander/Photo Disc/Getty
		<i>C. perfringens</i> - causes gas gangrene (الغرغرينا) and food poisoning.
		<i>C. acetobutylicum</i> - manufacture of butanol
Desulfotomaculum	Peritrichous or polar flagella	is the ONLY bacteria that reduces sulfate and sulfite to hydrogen sulfide (H_2S) during <u>anaerobic</u> respiration.  © J. R. McLaughlin/Photo Disc/Getty
Heliobacterium	Gliding motility	- photoheterotrophic: photosynthetic bacteria (have a bacteriochlorophyll gene) - stains gram-negative but in electron micrographs is seen to have a gram-positive cell wall. هاي البكتيريا من شواذ قاعدة الـ gram staining لأنها بتعطي نتيجة الـ gram negative ولكن لما يتم وضعها تحت الـ electron microscope بيظهر الـ cell wall تبع الـ gram positive.
Viellonella	Nonmotile Cocci	gram-negative but placed in order Clostridiales (mentioned in the previous chapter)

3. Sporolactobacillus

is a lactose fermenter spore former that ferments lactose under anaerobic conditions and these are not used in dairy products because they form endospores and cause food spoilage later on.

B. Nonsporeformers

Are divided into Regular and Irrgular

i. Regular

1. Order Lactobacillales

- also called lactic acid bacteria
- morphologically diverse
 - nonsporing
 - usually nonmotile
- ferment sugars for energy
 - lack cytochromes
 - fastidious*

**fastidious microorganism is used in microbiology to describe microorganisms that will grow only if special nutrients are present in their culture medium.*

- contains several important genera
- largest genus - Lactobacillus

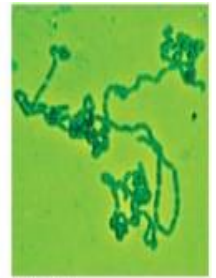
- bacilli but some are coccobacillus: very difficult to distinguish under the microscope
- grow optimally in slightly acidic conditions (pH 4.5 to 6.4)



(a) Lactobacillus acidophilus



(b) L. lactis
a. R. Arthur M. Sengmeier/Visuale Unlimited
© George J. Wiesner/Visuale Unlimited



(c) L. bulgaricus

- carry out either homolactic fermentation (via **glycolytic pathway**) or heterolactic fermentation (via **pentose phosphate pathway**)

→ Genus Lactobacillus:

- widely distributed in nature
 - on plant surfaces
 - in dairy products, meat, water, sewage, beer, fruits, and other materials
 - normal flora of mouth, intestinal tract, and vagina
 - usually not pathogenic
- aerotolerant (lives equally well with or without oxygen) and this is a huge advantage in dairy production

بتكمن اهمية كونها aerotolerant بصناعة منتجات الحليب حيث انو لما يتم خلطها مع الحليب لصناعة اللبن مثلا.. رح يتحول الحليب الي فوق والي تحت كلو اللبن .. لو انها كانت مثلا aerobic، كان بس تحول الحليب اللي من فوق اللبن واللي تحت ضل حليب



- Importance of lactobacilli
 - fermented vegetable products (sauerkraut, pickles, and silage)
 - fermented beverages (beer, wine, juices)
 - sour dough bread
 - Swiss cheese and other hard cheeses
(We have many kinds of cheese depending on what starter culture we choose (what kind of bacteria))
 - yogurt
 - sausages
 - spoilage of beer, milk, and meat



2. *Listeria*

- short rods
 - motile by peritrichous flagella
- aerobic or facultative
 - catalase positive
- common in decaying matter
- e.g., *L. monocytogenes* - listeriosis
 - food-borne disease
- causes β -hemolysis to blood

3. Genus *Caryophanon*

- disk-shaped cells
 - *Caryophanon* individual bacterium is disk shaped and they stack side by side forming a rod
- normal habitat is cow dung
- strict aerobic
- causes hypersensitivity and allergy

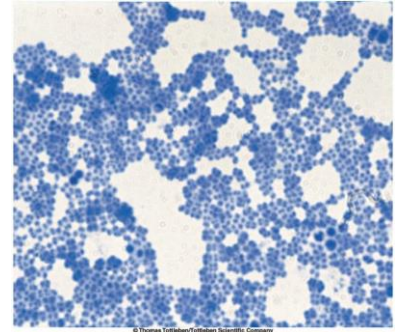


From J.G. Hall (Ed.), The Sauer-Bergers Manual of Systematic Bacteriology, Vol. 2, © 1980 Williams and Wilkins Co., Baltimore

ii. Irregular

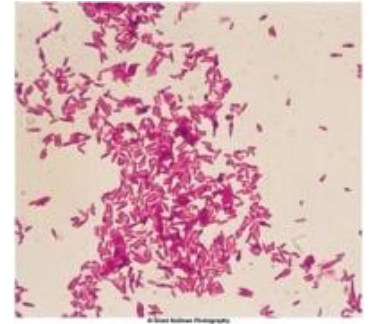
1. *Arthrobacter*

- aerobic, catalase-positive rods
- respiratory metabolism
- lysine in peptidoglycan
- found mainly in soil and has the ability to breakdown most herbicides and pesticides.
- Its irregular because it has the rod to coccus growth cycle, which grows as bacilli and turns to cocci once reached the stationary phase due to nutrient deprivation.



2. Suborder *Corynebacterineae*

- has seven families with many known genera such as
 - *Corynebacterium*
 - *Mycobacterium*
 - *Nocardia*

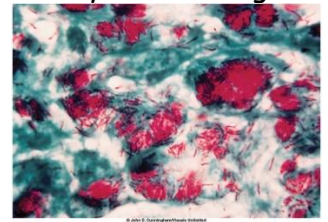


Genus	Oxygen requirement	slides	Additional notes
<i>Corynebacterium</i>	Aerobic or facultatively anerobic	• only genus in Family <i>Corynebacteriaceae</i> • some are harmless soil and water saprophytes • many are animal and human pathogens – e.g., <i>C. diphtheriae</i> - diphtheria • after snapping division bacteria often remain partially attached resulting in palisade arrangements of cells منحكيها انها irregular لانها بس تنقسم بال binary fission الخلية الجديدة رح تضل ملتصقة بالاولى	• <i>Corynebacterium diphtheriae</i> secretes a powerful toxin* as powerful as the toxin secreted from <i>Clostridium botulinum</i>. • <i>Corinebacterium diphtheria</i> produces a pseudomembrane to protect itself. This membrane gets bigger and block the airway causing suffocation and it secretes a toxin on cells and kills them 

Mycobacterium	aerobic and catalase positive	• in family <i>Mycobacteriaceae</i> • straight or slightly curved rods that sometimes branch or form filaments • contain waxes with 60-90 carbon mycolic acids • acid-fast - basic fuchsin dye cannot be removed from cell by acid alcohol treatment • Important species of <i>Mycobacterium</i> - <i>M. bovis</i> - tuberculosis in cattle and other ruminants - <i>M. tuberculosis</i> - tuberculosis (السل) in humans - <i>M. leprae</i> - leprosy	→ <i>Mycobacterium</i> is normally like filaments in shape. if disturbed, they are fragmented to form rods and cocci bodies. → <i>Mycobacteria</i> can avoid being digested by phagocytic cells because they have a waxy cell wall (high lipid content and mycolic acid) and that cell wall protects it from the phagocytic enzymes since their lysosomes lack lipases so it lives inside the phagocytic cell and around with it الـ <i>Mycobacteria</i> ما يتم هضمها من قِبل الـ <i>phagocytic cell</i> لانيو الجدار الخلوي تبعها عبارة عن مادة شمعية يتكون من <i>mycolic acid</i> وهو نوع من الـ <i>lipids</i> ف لما الـ <i>lysosomes</i> تبعت الـ <i>phagocytic cell</i> ما يكون فيها <i>lipases</i> (الانزيمات الي بتحطم الـ <i>lipids</i>) فهي ببساطة ما رح تقدر تقتلها وحتضل عايشة معها • <i>Mycobacterium</i> is present in many humans as sub-clinical (no symptoms shown on the person infected) and most infections in humans cause in asymptomatic carriers (infected but not showing any symptoms).
---------------	-------------------------------	---	--

→ Two major causes to turn the asymptomatic carriers to symptomatic is when infected people have mental and physical stresses

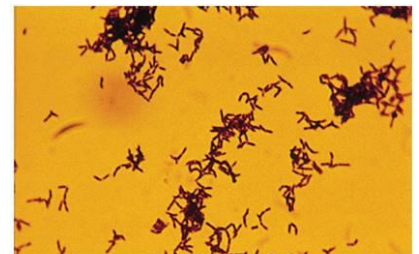
- Mycobacteria are slow growers, to grow them in lab you need to incubate them for 2-40 days on solid agar



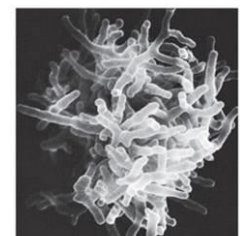
* The most important point when treating a disease that is caused by a bacteria that produces a toxin is that we should neutralize the toxin using antitoxin and then we use antibiotic to kill the bacteria because the problem is not with the bacteria itself, it's with the toxin it produces

3. Genus *Actinomyces*

- straight or slightly curved rods and slender filaments with true branching
 - may have swollen, clubbed, or clavate ends
- facultative or obligate aerobes
 - require CO_2 for best growth
- normal flora of mucosal surfaces (especially oral cavity) of humans and other animals
 - e.g., *A. bovis* - lumpy jaw in cattle
 - e.g., *A. israeli* - most important human pathogen
 - actinomycoses - ocular disease and periodontal disease in humans



(a)



(b)

The End

(~▽~)~ ~(▽~)~